

LLM-Based Extraction of Pharmacogenomic Gene-Drug Associations from PharmGKB Text

A Feature-Fusion Approach for Warfarin Dose Prediction

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Technical Report

Abstract

Abstract. Most pharmacogenomic prediction models work off curated, structured data — genotype, demographics, clinical variables. But a lot of the actual evidence linking genes to drug outcomes lives in plain text: PharmGKB annotations, literature excerpts, evidence sentences that never get turned into a clean feature column. This report looks at whether a fine-tuned LLM can pull that association signal directly out of the text, and whether feeding it into an existing structured model actually helps on a real downstream task. We fine-tune Qwen2.5-3B-Instruct with LoRA to classify gene-drug association status from PharmGKB text, then fuse its output with structured features and use XGBoost to predict warfarin dosing on IWPC data. On a held-out set of 1,655 gene-drug pairs, 1,439 had clean associated/not-associated labels and were used for evaluation; the model hit 84.09% accuracy on that set ($F1 = 85.02\%$). On the downstream warfarin task — 658 records from 333 patients, split so no patient leaks across train/test — the fused model reached $R^2 = 0.70$, $MAE = 7.0$ mg/week, $RMSE = 10.0$ mg/week. We're upfront throughout about what this does and doesn't show, including that we haven't yet run a structured-only baseline to isolate exactly how much the LLM features are contributing.

1 Introduction

Here's the gap this project is poking at: pharmacogenomic dosing models are usually built on structured features — a patient's genotype, some clinical variables, maybe demographics — because that's what's easy to plug into a model like XGBoost. But most of the actual evidence for why a gene affects a drug response lives in text. PharmGKB curates a lot of this into a database, but the curation itself is manual, and the annotation sentences underneath the curated labels carry more nuance than the structured summary captures.

So the question here is simple: can an LLM, fine-tuned specifically for this, read that text and pull out reliable association signal — and if it can, does adding that signal to an existing structured model actually move the needle on something real? We test this on warfarin dosing, since it's one of the most well-studied pharmacogenomic dosing problems out there and has a public dataset (IWPC) to work with.

One thing worth being clear about upfront: Qwen's job in this pipeline is extraction, full stop. It reads a sentence and says "these two things are associated" or "they're not." It doesn't recommend a drug, and it doesn't predict a dose. That's XGBoost's job, downstream, using Qwen's output as one input among several.

What's actually in this report:

1. A Qwen2.5-3B-Instruct model, fine-tuned with LoRA, that classifies gene-drug association status from PharmGKB text — tested on a held-out split built so no gene-drug pair leaks between train and test.
2. A simple way of turning that model's output into numerical features that can sit next to genotype data.
3. A real test of whether those features help on warfarin dose prediction, using IWPC data, split by patient so nobody's records show up in both train and test.
4. An honest list of what we didn't check yet — mainly, we haven't run the structured-only version of the dosing model, so we can't say exactly how much of the result is the LLM features doing work.

To be clear about what kind of contribution this is: it's not a new way of doing feature fusion. That's been done before, in different forms. What this report actually contributes is a specific, working version of the idea — Qwen reading PharmGKB text, feeding structured features, landing on a real downstream warfarin prediction task, evaluated honestly.

2 Related Work

2.1 Pulling structured info out of pharmacogenomic text

This isn't the first attempt at getting machines to read pharmacogenomic literature. PGxMine, for instance, built extraction pipelines aimed at PharmGKB-style annotations — mostly focused on pulling out entities and relationships from the literature at scale, rather than feeding that output into a downstream predictive model like we do here.

2.2 What PharmGKB actually contains

PharmGKB's curated relationships are useful, but they're a simplification. The evidence sentences behind each curated label usually say more — which variant is involved, what phenotype shows up, which direction the effect goes, and whether it's a pharmacokinetic (PK), pharmacodynamic (PD), efficacy, toxicity, or metabolism relationship. In principle, extracting straight from those sentences could capture more than a binary "associated / not associated" label ever will. That said — and this matters — the task we actually run in this report is the simpler binary version, not the richer one. More on that in Section 4.

2.3 Warfarin dosing, specifically

The IWPC did the groundwork here — establishing that genes like *CYP2C9* and *VKORC1* matter a lot for predicting how much warfarin a given patient needs, on top of the usual clinical and demographic variables. Since then, people have thrown various ML methods at this problem, including tree-based models like XGBoost, using IWPC-style data.

2.4 Combining LLM output with structured features

This general move — take something an LLM pulled out of text, turn it into a feature, combine it with structured data — isn't new either. There's work using LLM-generated, evidence-grounded pharmacogenomic recommendations straight from literature. There's also LLM-DDI, which fuses LLM embeddings with graph neural networks to improve drug-drug interaction prediction. So to

be clear: fusing text-derived and structured signal is a known move. We're not claiming to have invented it.

2.5 Where this report actually sits

Put together: the extraction side builds on PGxMine-style work, the domain knowledge comes from PharmGKB and the IWPC warfarin literature, and the fusion idea has precedent elsewhere. What this report does is connect all of that into one specific, tested pipeline — Qwen doing the reading, XGBoost doing the predicting, warfarin dosing as the real downstream task it gets evaluated on. That connection, evaluated honestly, is the contribution.

3 Data

For extraction. We pulled PharmGKB gene-drug annotation text covering 11,035 gene-drug pairs. The split — 7,725 training, 1,655 validation, 1,655 test — is done *by pair*, not by sentence. That matters: it means a given gene-drug pair only ever shows up in one split, so the model can't just memorize a pair it already saw during training and get an easy win at test time.

For the downstream dosing task. We used IWPC data: 658 records from 333 patients. Here the split is done by patient, not by record — so no single person's data shows up in both the training set and the test set.

4 Method

4.1 What "association" actually means here

Worth spelling this out clearly, because it's easy to overstate what this model is doing. Qwen is being used purely as an extraction tool — it reads a sentence, it doesn't diagnose anyone or recommend anything. The task itself is binary: does this PharmGKB text say the gene and drug are associated, or not? Passages where the gold label itself was ambiguous were left out of evaluation entirely — see Section 5.1 for the exact numbers.

And to be clear about the ceiling on this task: it's a coarse yes/no classification. It doesn't try to figure out whether the relationship is pharmacokinetic, pharmacodynamic, about efficacy, toxicity, or metabolism. That's a different, harder task — this report doesn't attempt it, and the result shouldn't be read as if it did.

4.2 Fine-tuning setup

We fine-tuned Qwen2.5-3B-Instruct with LoRA. Table 1 has the exact configuration.

4.3 The structured baseline

This part isn't new work — it's a structured XGBoost drug-recommendation model built on genotype, demographic, and clinical features, from earlier work (Sakhalkar et al., prior work). We're reusing it as the backbone this new pipeline builds on top of.

4.4 How the two pieces connect

Qwen's predictions on each passage get turned into simple numerical features — whether an association was flagged, how confident the model was, that kind of thing — and those get concatenated onto the existing structured feature set. That combined feature vector is what actually

Table 1: LoRA fine-tuning configuration.

Hyperparameter	Value
LoRA rank (r)	16
LoRA alpha	32
LoRA dropout	0.05
Target modules	q_proj, k_proj, v_proj, o_proj
Quantization	4-bit NF4
Learning rate	1×10^{-4}
Training epochs	2
Per-device batch size	2
Gradient accumulation steps	8

trains the downstream XGBoost model for warfarin dosing. To say it plainly one more time: Qwen extracts, XGBoost predicts. Different jobs.

5 Results

5.1 How well the extraction model actually did

Out of the 1,655 held-out gene-drug pairs, 1,439 had clean, unambiguous gold labels and were used for evaluation; the remaining 216 pairs were left out because their gold association label was ambiguous or missing. Table 2 shows how Qwen did on that 1,439-instance evaluation set.

Table 2: Qwen’s binary association classification performance (1,439-instance evaluation set).

Metric	Value
Accuracy	84.09%
Precision	83.44%
Recall	86.67%
F1	85.02%

Table 3: Confusion matrix, same evaluation set.

	Predicted Negative	Predicted Positive
Actual Negative	560 (TN)	129 (FP)
Actual Positive	100 (FN)	650 (TP)

5.2 Does it actually help downstream: warfarin dosing

This is the real test — does the extracted evidence do anything useful once it’s part of a real prediction task? We ran the fused pipeline (structured features + Qwen-derived features) on warfarin dose prediction using IWPC data, held out by patient. Results in Table 4.

Here’s the honest caveat: we haven’t run the same task with structured features *only*, so we can’t yet say how much of this $R^2 = 0.70$ is coming from the Qwen features versus what the structured features alone would have gotten anyway. Published IWPC-style dosing algorithms

Table 4: Fused model performance on warfarin dose prediction.

Metric	Value
R^2	0.70
MAE	7.0 mg/week
RMSE	10.0 mg/week

tend to land in a broadly comparable R^2 range on similar cohorts, which is useful context, but it’s not a substitute for our own ablation — that’s next on the list. Also worth repeating: this number describes the downstream XGBoost regression, not Qwen’s classification accuracy — those are two different numbers for two different jobs, reported separately.

6 Discussion and Limitations

A few things worth saying plainly. The 84.09% figure is extraction accuracy on a coarse binary task — it is *not* a clinical accuracy number, and it shouldn’t be read as if the model is doing full relation extraction (it isn’t distinguishing PK from PD from toxicity from metabolism — just yes/no association).

The warfarin result is real and honestly promising, but 658 records from 333 patients is a modest sample next to the scale usually used in dosing-algorithm papers, so it’s worth being cautious about how far this generalizes. And again — without the structured-only ablation, we genuinely don’t know yet how much of the improvement is the LLM’s doing.

This isn’t a clinical tool and it’s not being framed as one. Qwen doesn’t make treatment decisions — it extracts evidence that becomes one input, among several, to a separate model. If a system like this were ever used in a real clinical context, a false positive on a gene-drug association would carry real downstream risk, which is exactly why this stays framed as a research pipeline, not a deployable product.

7 Future Work

A few clear next steps: test on a bigger, more diverse pharmacogenomic dataset so the results generalize further; run the structured-only ablation so we can actually attribute the warfarin result properly; push extraction beyond binary association into full relation typing (PK/PD, efficacy, toxicity, metabolism); and try larger models or different PEFT setups to see if that helps with the harder cases — negation, conditional statements, multi-relation sentences — and cuts down how many instances end up ambiguous in the first place.

8 Conclusion

This report builds a pipeline where Qwen2.5-3B-Instruct reads PharmGKB text and classifies gene-drug association status, and where that output feeds into an XGBoost model for warfarin dose prediction alongside structured genotype and clinical features. On the 1,439-instance evaluation set (out of 1,655 held-out pairs, 216 excluded as ambiguous), the extraction model reached 84.09% accuracy. The full downstream pipeline reached $R^2 = 0.70$ on a patient-level held-out warfarin dosing test. We’ve tried to be upfront about the gaps — mainly the missing structured-only ablation — and this sits as one specific, working example within a broader, active line of research on using LLMs to extract pharmacogenomic evidence and fuse it with structured prediction, rather

than as a claim to have invented that idea.